Project Proposal: Prompt-engineering-based Hallucination Estimation for large language models

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Background and Motivation

In recent years, large language models (LLMs) have demonstrated unprecedented capabilities in numerous natural language processing (NLP) tasks, including reasoning, generation, and knowledge-intensive applications. Despite this progress, a persistent challenge remains: hallucinations. Hallucination occurs when model-generated content appears fluent and plausible, but is inconsistent with user input, context, or factual knowledge (Zhang et al., 2025). This phenomenon is particularly serious in real-world scenarios such as question-answering, summarization, and medical consultation, potentially leading to the dissemination of misleading information, reducing reliability, and even posing real risks (Zhang et al., 2025).

To address this issue, the academic community has begun systematically studying the definition and classification of hallucinations in large language models (LLMs). For example, Zhang et al. (2025) categorized hallucinations into Input-conflicting Hallucination, Context-conflicting Hallucination, and Fact-conflicting Hallucination, and proposed corresponding evaluation benchmarks to more accurately identify model errors in different dimensions. Huang et al. (2025) further distinguished between Faithfulness Hallucinations and Factual Hallucinations, noting that these two types of hallucinations have different manifestations and mitigation challenges in scenarios such as Retrieval-Augmented Generation (RAG). Meanwhile, Ji et al. (2023) categorized hallucinations into Intrinsic Hallucinations and Extrinsic Hallucinations, further emphasizing that hallucinations exist not only in LLMs but also in a wide range of natural language generation (NLG) tasks, including summarization, generative question answering, data-to-text generation, machine translation, and dialogue generation, highlighting the universal challenge of hallucinations in language generation systems.

From a causal perspective, hallucinations often arise from multiple factors. On the one hand, training data inevitably contains noise, fabrication, bias, and outdated information, resulting in misleading and erroneous output (Zhang et al., 2025); On the other hand, strategies such as top-k sampling during decoding can increase the probability of hallucinations, causing the model to generate text that appears plausible but is essentially erroneous (Ji et al., 2023). In addition, when models lack sufficient up-to-date knowledge and data coverage in knowledge-intensive domains such as medicine, law, and finance, they are more likely to produce fabricated answers (Niu et al., 2024). The risks posed by this phenomenon are particularly pronounced in high-risk scenarios. For example, in medical question-answering systems, hallucinations can lead to misdiagnoses or incorrect treatment recommendations (Niu et al., 2024). Furthermore, hallucinations not only undermine user trust in LLMs but also limit their safe deployment in production environments (Tonmoy et al., 2024). Therefore, hallucinations are not only a challenge for academic research but also a bottleneck that urgently needs to be addressed in the practical deployment of LLMs.

Addressing hallucinations is crucial for improving the reliability and trustworthiness of LLMs. On the one hand, reducing hallucinations can enhance the application value of LLMs in high-risk scenarios (such as healthcare, law, education, and finance) (Niu et al., 2024). On the other hand, hallucination detection and mitigation methods, including those based on uncertainty estimation (Xiong et al., 2024), retrieval enhancement (Niu et al., 2024), and prompt-engineering (Tonmoy et al., 2024), can not only improve model performance, but also provide users with a more transparent interactive experience. However, existing research is still insufficient to combine detection and mitigation, and many works focus on detecting hallucinations or proposing mitigation methods but lack a systematic closed-loop mechanism from detection to optimization (Tonmoy et al., 2024).

Therefore, the significance of this project lies in the detection, estimate, and resolution of hallucinations in large language models through prompt-engineering, improving the factual consistency and answer accuracy of LLMs, and establishing a lightweight, yet effective framework for hallucination detection and mitigation. This framework not only provides new insights for academic research, but also provides more reliable LLM support for practical applications. Meanwhile, I was motivated to choose this topic primarily because, through previous courses and projects, I have accumulated experience with AI methods and large-scale model applications. This has given me a deep understanding of the reliability deficiencies of large language models in real-world scenarios, and I hope to explore how to improve their trustworthiness. Furthermore, this research direction aligns closely with my future academic plans to pursue further studies in artificial intelligence and data science, so I hope to gain cutting-edge research and practical experience through this project.

Aim(s) and Objectives

This project aims to detect and estimate the probability of hallucinations in model outputs and then use Prompt-engineering to effectively mitigate and correct hallucinations, thereby improving the reliability and factual consistency of large language models in practical applications.

The key objectives of this research are

- **Literature Review**

  1. Systematically review the definition and classification of large language model hallucinations.
  2. Analyze the main causes of hallucinations.
  3. Summarize existing detection and mitigation methods, comparing their strengths and limitations.
  4. Identify research gaps and propose a unique approach for this project: the lack of a closed-loop mechanism that combines detection and mitigation.

- **Datasets**
  To ensure breadth and reproducibility, this project will evaluate the methods I subsequently devised on multiple tasks and datasets. Current potential datasets include:

  1. **Open-domain QA:** TruthfulQA and NaturalQuestions
  2. **Retrieval-augmented QA (RAG)**
  3. **Summarization**

- **Models, Resources, and Risk Controls**

  1. **Model Selection**
     This project will primarily utilize open-source LLMs such as LLaMA-3 or others.

2. **Resource Budget**

   To ensure efficiency and cost control, token and latency budgets will be set.

3. **Risk Controls**

   In high-risk or low-confidence situations, the system will apply a default abstention or clarification strategy, refusing to answer or requesting clarification rather than generating unreliable responses.

- **Detection Mechanism**

  1. Research and select suitable open-source hallucination detection tools or libraries for preliminary hallucination identification and verification of LLM outputs.
  2. Further design and implementation of custom detection mechanisms are possible, if necessary.

- **Mitigation Strategy**

  Based on the results of the detection phase, this project will explore and selectively adopt a range of prompt-engineering strategies to optimize model outputs. Potential approaches include, but are not limited to:

  1. Prompt Rephrasing

     Guide the model to give more factually consistent answers.

  2. Context-Augmented Prompting

     Provide more external knowledge or retrieve information to the model.

  3. Chain-of-Thought Prompting

     Reduce reasoning-related hallucinations.

  4. One-to-Many Prompting

     Generate multiple different candidate answers to the same question and select the most reliable output through consensus or majority voting.

  5. Many-to-Many Prompting

     Multiple related questions are cross-correlated with multiple candidate answers, and are verified and corrected using contextual consistency and semantic alignment.

6. Innovative Strategy

   Based on the above, independently design new mitigation strategies. For example, a closed-loop pipeline (estimate → mitigate → re-evaluate) is conceptually proposed and may be implemented depending on pilot study outcomes

- **Detection Evaluation Process**

  1. Use hallucination detection methods to evaluate the model's raw output and quantify the incidence of hallucinations.
  2. Apply prompt-engineering to the same task and dataset for optimization, and again use the detection mechanism to assess the hallucination rate.
  3. Compare the before-and-after detection results, using this as the core metric for measuring the effectiveness of prompt-engineering.
  4. Conduct baseline experiments comparing our approach with existing non-closed-loop methods to evaluate whether the proposed closed-loop framework provides measurable improvements in hallucination detection and mitigation.
  5. Conduct ablation studies by selectively removing or disabling key components to quantify the contribution of each module to overall performance.
  6. Conduct significance tests to ensure that the performance improvement is meaningful.

  In addition, future work may extend the evaluation to more degrees.

Work Plan

To achieve the project objectives, a corresponding time frame and specific tasks have been outlined in Figure 1. The project preparation phase will take place between late September and early November 2025, primarily encompassing project proposals, submission of ethical review forms, and selection of large prediction models and datasets. A comprehensive literature review will be conducted from early October to mid-December, aiming to investigate the definition, classification, causes, and mitigation approaches for hallucinations related to LLM. From late October to November, the detection mechanisms will be explored through the selection of hallucination detection tools and test datasets.

A midterm report will be drafted in December 2025 and submitted by December 16.

From January to February 2026, the project will focus on delivering a minimum viable closed-loop prototype, including implementation of an initial detection-mitigation-reassessment cycle and preliminary error analysis to evaluate system performance. From February to March 2026, the focus will shift to rapid engineering strategy development, encompassing both foundational and innovative strategies to mitigate hallucinations related to inference. From March to April 2026, the project will enter the evaluation and benchmarking phase, conducting efficiency analysis, ablation experiments, and performance evaluation across all selected tasks and datasets. The dissertation will be completed and polished by April 2026, followed by the production of a demonstration video and preparation for a QA session in late April to May 2026.

This structured schedule ensures steady progress across all phases, providing ample time for iteration, validation, and documentation of the closed-loop hallucination mitigation framework.

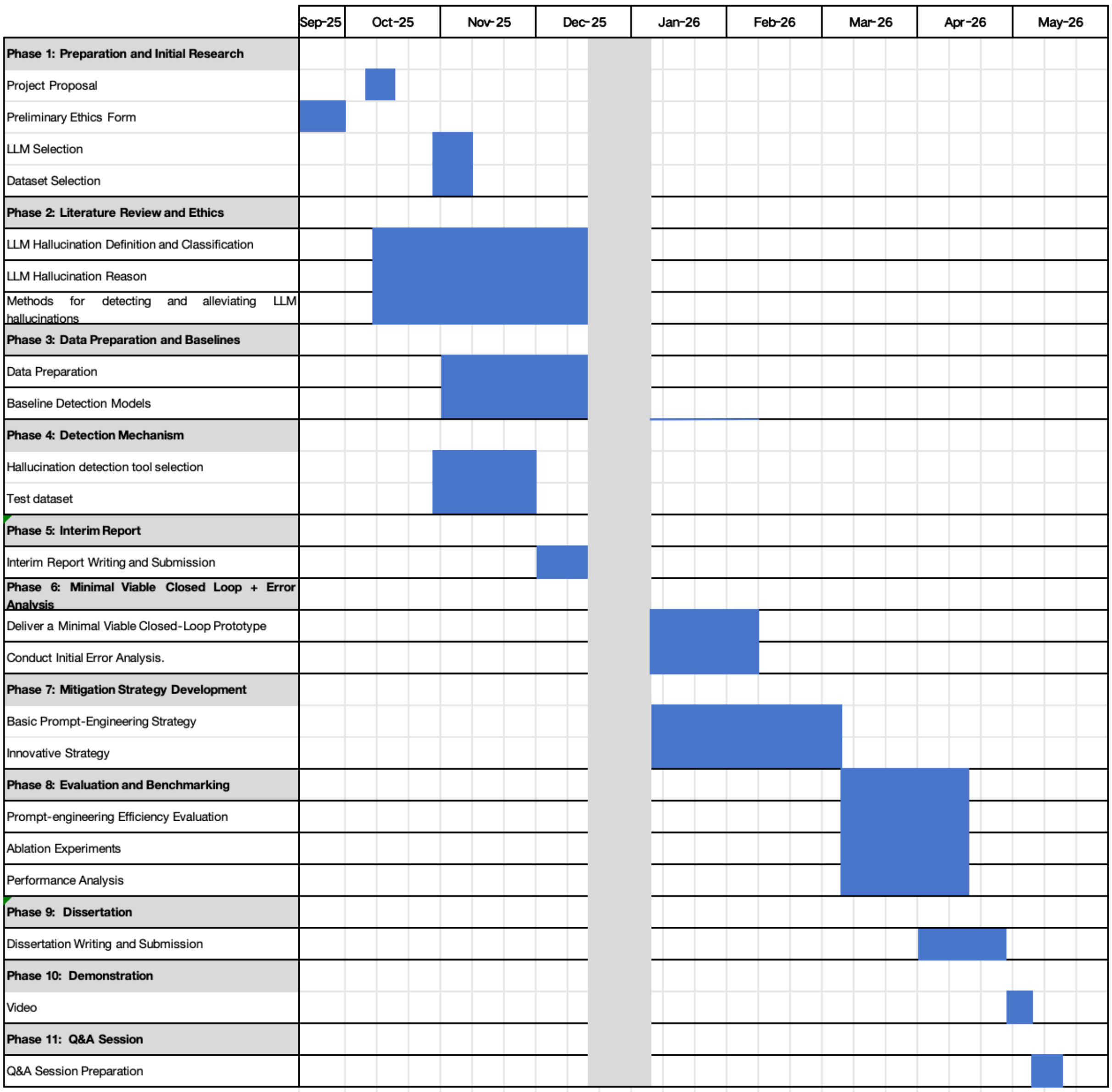

Figure 1: *Project Milestones and Time Plan*

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